Try It 5.28

Solve the system by substitution. $\begin{cases} 2x - y = 1 \\ y = -3x - 6 \end{cases}$

If the equations are given in standard form, we'll need to start by solving for one of the variables. In this next example, we'll solve the first equation for y.

Example 5.15

Solve the system by substitution. $\begin{cases} 3x + y = 5 \\ 2x + 4y = -10 \end{cases}$

Solution

We need to solve one equation for one variable. Then we will substitute that expression into the other equation.

Solve for y. Substitute into the other equation.	$3x + y = 5$ $y = -3x + 5$ $2x + 4y = -10$
Replace the y with $-3x + 5$.	$2x + 4(-3x + 5) = -10$
Solve the resulting equation for x.	$2x - 12x + 20 = -10$
	$-10x + 20 = -10$ $-10x = -30$
Substitute $x = 3$ into $3x + y = 5$ to find y.	$x = -3$ $3x + y = 5$
	$3(3) + y = 5$ $9 + y = 5$
The ordered pair is $(3, -4)$.	$y = -4$
Check the ordered pair in both equations: $3x + y = 5$ $3 \cdot 3 + (-4) \stackrel{?}{=} 5$ $9 - 4 \stackrel{?}{=} 5$ $5 = 5$ ✓ $2x + 4y = -10$ $2 \cdot 3 + 4(-4) = -10$ $6 - 16 \stackrel{?}{=} -10$ $-10 = -10$ ✓	
	The solution is $(3, -4)$.